Enhanced Aircraft Detection in Compressed Remote Sensing Images Using CMSFF-YOLOv8

Sharan Thapa, Yuqi Han, Baojun Zhao, and Senlin Luo

***Abstract*—Object detection is an important part of remote sensing image analysis to accurately identify key features in Earth observation data while minimizing resource usage. However, challenges such as multiscale object representation and the identification of small-sized objects, particularly in compressed images, persist. To tackle this problem, we propose a state-of-the-art detector called compressed multiscale feature fusion YOLOv8 (CMSFF-YOLOv8) that employs a four-channel discrete wavelet transform (DWT) coefficient image representation for preprocessing. This method processes the four coefficients of the three-level discrete wavelet transform (3-DWT) result, arranging them into a four-channel stacked image representation that retains the important spatial–frequency characteristics of the information. The improved preprocessing architecture design employs frequency-domain filtering (FDF) to retain important edge information, adaptive histogram equalization (AHE) to adjust the intensity distribution locally, and multiscale feature transfer (FT) to enable the extraction of important edge features at different scales. Squeeze-and-excitation convolution (SEConv) for channel-wise attention and multigranularity enhanced feature aggregation (MGEFA) for multilevel feature extraction of DWT coefficients are integrated into the backbone of the CMSFF-YOLOv8 architecture, enabling the selective prioritization of structural and edge features in complex settings. Spatial pyramid pooling fast (SPPF) combined with shallow feature fusion networks (SFFNs) enhances small object detection. A custom neck design with upsampling facilitates multiscale feature fusion (MSFF). The result is mapped back to the LL3 channel in the compressed domain, reducing computational complexity while retaining critical global context. The proposed CMSFF-YOLOv8 detector, combined with a four-channel image representation and advanced preprocessing, demonstrates an efficient and accurate solution for object identification in compressed-domain remote sensing applications, achieving an *F*1-score of 0.8473, a recall of 0.82165, a precision of 0.8746, and mean average precision (mAP) values of 0.90895 at intersection over union (IoU) = 0.5 and 0.60063 at IoU = 0.75 on the HRPlanesV2 dataset.**

***Index Terms*—Feature fusion, feature transfer (FT), four-channel image representation, multilevel discrete wavelet transform (DWT), small object detection.**

Received 27 June 2025; revised 19 August 2025 and 11 October 2025; accepted 15 October 2025. Date of publication 20 October 2025; date of current version 10 November 2025. *(Corresponding authors: Sharan Thapa; Yuqi Han.)*

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Digital Object Identifier 10.1109/TGRS.2025.3623996

I. Introduction

THE aircraft identification in remote sensing satellite images is crucial for applications that require accurate and timely detection to ensure the safety and security of airspace [1], [2] and airports [3]. As remote sensing technology advances, the volume of satellite data increases, and these delays become more pronounced. Traditional identification methods fail to address this complexity due to their inability to examine compressed data directly. This leads to delays and inefficiencies. Additionally, manual feature extraction is challenging due to variations in aircraft size, position, orientation, and complex backgrounds. These challenges [4] highlight the need for advanced detection methods that enhance accuracy and reliability. This method [5] demonstrates improved classification accuracy through a deep feature fusion network (DFFN). Various detection approaches [6], [7] have been proposed, each tailored to specific applications. Real-time systems offer low-latency processing. Some deep learning models may have high inference times. Most fast detection methods [8], [9], [10], [11], [12] prefer object detection without compression techniques, as compression and decompression introduce delays and require significant resources and bandwidth.

The multilevel discrete wavelet transform (DWT) [13], [14] reduces an image's size without destroying important information. This compression method breaks down an image into several frequency coefficient levels. It contains the broad structural elements and the fine details (edges, textures). The low-frequency components (LL) are usually the most important for retaining visual quality, while the high-frequency components (HL, LH, and HH) contain finer details, such as edges and small textures. In a three-level DWT (3-DWT), the LL subband from the first level (LL1) is decomposed into LL2 and high-frequency subbands (LH2, HL2, and HH2). Then, LL2 is further decomposed into LL3, LH3, HL3, and HH3. The low-frequency subbands (especially LL3 in the third level) retain the essential visual information and global features of the image, making them suitable for structural representation. In contrast, the high-frequency subbands (LH, HL, and HH at each level) capture localized variations and fine textures, which are critical for identifying small and detailed objects, such as aircraft in remote sensing imagery. These components can be compressed more efficiently by reducing their size while retaining the essential features necessary for high-quality reconstruction [15], [16]. Feature transfer (FT) in image processing refers to extracting important features from part of an image for the decision-making process in the

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identification of the target [17], [18], [19]. In multilevel DWT, important edge and texture features can be passed through different layers of decomposition to achieve good compression ratios without significant loss of visual information. Increasing attention has been directed toward leveraging compressed-domain features within deep learning frameworks for object detection.

The system is more resilient to noise and other distortions because a multilevel decomposition-based methodology ensures that the image's high- and low-frequency signals are used for feature extraction. Frequency-domain filtering (FDF) [18], [20], [21] is a useful method for enhancing image quality and capturing minute details, such as textures and edges. It highlights the edges of the image. We are applying it to extract detailed edge information in the higher frequency coefficient. Another key challenge in satellite image processing is the difference in illumination and contrast across different areas of the image, which can be detrimental to important features. Adaptive histogram equalization (AHE) [22], [23] addresses these issues by enhancing contrast locally, making it particularly useful for detecting small objects. Unlike conventional histogram equalization, AHE dynamically adjusts contrast based on local intensity variations, resulting in more effective feature enhancement. By locally modifying the intensity distribution, AHE improves the visibility of small targets, especially in regions with low contrast and poor illumination. It is very helpful for our applications to improve image quality and enhance the accurate edge recognition of partially visible aircraft under cloudy conditions as well. For real-time performance, it is important to use simpler models and smart algorithms to ensure that the system responds quickly and does not use too many resources. Additionally, at low compression ratios, the FT approach in multilevel DWT maintains important target information, such as the edge information of the aircraft.

The you only look once (YOLO) architecture [24], [25], [26], [27], [28] is popular nowadays for detecting objects quickly and accurately; it can be modified to handle compressed-domain data, further improving detection performance. The YOLOv8 can take the four coefficients of the DWT as a four-channel stacked image input by changing the YOLOv8 input image downloader. YOLOv8 focuses on architectural improvements, such as the cross-stage partial (CSP) Darknet53 backbone and an anchor-free identification head mechanism to balance speed and accuracy. When choosing one of these architectures, one should take into account the computational resources, specific application requirements, and the difficulty of the current real-time identification tasks. Fast detection YOLO architectures [8], [9], [10], [11], [12] and the proposed compressed multiscale feature fusion YOLOv8 (CMSFF-YOLOv8) detectors share foundational elements typical of YOLOv8 models, such as convolutional backbones and feature pyramid structures.

We proposed a preprocessing method that combines several simple and efficient techniques, incorporating enhanced modifications to achieve better results, such as 3-DWT for image compression, FDF for enhancing image quality and capturing fine details like textures and edges, wavelet FT to preserve important high-frequency features in higher decomposition levels, and AHE for adjusting contrast based on local intensity variations. These methods are applied as preprocessing steps for generating four-channel three-DWT coefficient image representations, aiming to improve accuracy, response speed, and overall performance. They enhance object identification in complex scenarios [29], [30] when objects are small, have low contrast, and are obscured by intricate backgrounds.

The CMSFF-YOLOv8 architecture is specifically designed to operate directly on four-channel DWT coefficient representations in the compressed domain. In contrast to previous approaches that depend on RGB inputs or necessitate full image reconstruction from compressed formats, our model directly utilizes both low- and high-frequency DWT subbands. The CMSFF-YOLOv8 architecture integrates several novel modules tailored for this compressed representation. The squeeze-and-excitation convolution (SEConv) module is designed to extract robust structural features from LL3 coefficients, while the multigranularity enhanced feature aggregation (MGEFA) module targets high-frequency components (LH3, HL3, and HH3) that are rich in edge details, enabling better detection of fine and small features. Additionally, the integration of shallow feature fusion networks (SFFNs) [31] with the spatial pyramid pooling fast (SPPF) module allows the model to capture fine-grained spatial details alongside multiscale contextual information. By concatenating shallow feature maps with the outputs of the SPPF module, the fusion combines multiscale contextual information with fine-grained details, thereby improving detection accuracy for tiny objects. The CMSFF-YOLOv8 neck serves as a bridge between the detection head and the backbone to detect objects of different size ranges, specifically P3–P5. It combines features from different layers. The proposed model has added a feature enhancement fusion module (FEFM) to the neck. The CMSFF-YOLOv8 detection head predicts bounding boxes and class probabilities. We have enhanced this component by integrating a feature fusion module (FFM). By incorporating the FEFM into the neck of CMSFF-YOLOv8 and the FFM into the detection head of CMSFF-YOLOv8, the proposed architecture achieves enhanced feature fusion capabilities.

The final output of the detector provides a boundary on the LL3 coefficient of the DWT that enables compressed-domain detection. The proposed method aims to address the challenges in satellite-based aircraft identification, monitoring, and airport management by eliminating the need for decompression at the receiver end [32]. Combining advanced feature extraction with multiscale feature mapping, along with previously mapped features and the current state, methods with small object detection algorithms [16] significantly reduce data volume, enhance processing speed, and optimize satellite resources. To enhance these techniques, standard feature extraction and object detection algorithms have been significantly improved. Integrating image compression-based preprocessing with small object detection in a single process can improve efficiency and reduce computational complexity. This serves as the core motivation and objective of the proposed CMSFF-YOLOv8 with enhanced preprocessing with a four-channel image representation framework. The major contributions of the proposed work are summarized as follows.

1) *Four-Channel Image Representation:* Development of a four-channel image framework that integrates enhanced

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AHE, multiscale FT, and FDF for improved feature representation in the compressed domain.

2) *Modified YOLOv8 Backbone:* Adaptation of the YOLOv8 architecture with a DWT coefficient-based backbone for compressed-domain processing, referencing high-frequency edge information from upper DWT bands to strengthen low-frequency decision features.
3) *Improved Multiscale Fusion and SFFN:* Enhancement of the SPPF module through the integration of an SFFN to capture fine-grained spatial details alongside multiscale contextual information, thereby strengthening small object detection performance.
4) *Optimized Real-Time Performance:* Modification of the YOLOv8 backbone, neck, and head to achieve real-time detection efficiency and high accuracy in the compressed domain.

II. Related Work

This section relates the connection between the proposed work and previous research, highlighting how our contributions enhance existing methods. The proposed model first focuses on image compression using DWT combined with feature enhancement methods. Second, it modifies the YOLOv8 baseline architecture to develop a compression-domain detector.

A. Image Compression With Enhanced Feature

The multilevel DWT is a hierarchical image compression method that breaks down an image into detail and approximation coefficients at various scales. Recent work such as WeConvene [33] integrates wavelet transforms into convolution and entropy coding to improve image compression. While their focus is on compression, the core idea of multiscale frequency decomposition aligns with our objective of enhancing object detection in the compressed domain. In contrast, our approach employs a 3-DWT combined with AHE and FDF preprocessing to generate rich spatial–frequency representations, preserving structural and edge information critical for small object detection. This highlights a novel application of DWT in downstream detection tasks beyond compression. As mentioned in [34], it is surprising to learn that high-frequency subband coefficients such as edge information are susceptible to information loss during multilevel decomposition, particularly in the HL, LH, and HH components that use downsampling to represent edges and textures. Previous baseline models are not capable of handling all complex scenarios involving small object identification. To address this issue, we implemented a sequence of techniques that support each other and enable efficient, accurate object detection directly on compressed-domain data while drastically reducing data volume. In the detection process, the edge of the higher frequency coefficient image is compared with the LL3 image coefficient for better image quality and accurate edge recognition. Recent advances in DWT have sought to mitigate this issue by carrying residual high-frequency information to higher levels, thereby improving detection in edge-sensitive applications such as satellite imagery.

B. Baseline YOLOv8

You only look once (YOLOv8) has been one of the series of models for object identification tasks because it can find a balance between speed and performance. Some other recent innovations, like YOLOv8 [35], have expanded detection abilities; they proposed a synthetic aperture radar (SAR) aircraft object detection algorithm with a gather-and-distribute (GD) machine, promoting multiscale feature fusion (MSFF) for detecting SAR aircraft in the complex backgrounds of SAR images. Additionally, when satellite image aircraft detection algorithms were extensively reviewed, YOLOv5 [36] proved to be a reliable choice that achieved high accuracy and adaptability across numerous imaging conditions. Some recent YOLO-based applications [8], [9], [10], [12] prefer detections of targets without compression, which require relatively huge resources, such as bandwidth. When choosing one of these architectures, one should take into account the computational resources, specific application requirements, and the difficulty of the current real-time identification tasks. The YOLOv8 architecture [24], [25], [26], [27], [28] is popular nowadays for detecting objects quickly and accurately; it can be modified to handle compressed-domain data, further improving detection performance. However, the combination of DWT with YOLOv8-type models and the use of the DWT for small object detection tasks are still limited. YOLO-based detection has also been extended to nonfully focused imaging domains, as in you only look once with reparameterized convolution (YOLO-RC) [37], which targets SAR ship detection in the range-compressed domain by leveraging amplitude gradient and geometric scale characteristics, while YOLO-extract [11] enhances YOLOv5 to improve small object detection and reduce computational costs. While their focus lies in optimizing YOLOv5's architecture for better accuracy and efficiency in natural and remote sensing images, our approach modifies YOLOv8 using multiscale DWT-based feature fusion to enhance detection performance in compressed imagery. Both works share the goal of improving the detection of small, dense targets in complex scenes but differ in the domains and structural modifications applied.

C. CMSFF-YOLOv8

Leveraging recent advances in object identification, we propose a variant of the YOLOv8-based [8] method with lightweight features for precise aircraft identification from satellite imagery, which is an improvement compared to existing state-of-the-art models. The proposed CMSFF-YOLOv8 architecture introduces modules such as SEConv for enhanced LL3 feature extraction, MGEFA to enhance high-frequency coefficient feature extraction, and attention mechanisms for DWT coefficients. SFFNs [31] enhance small object detection by integrating fine-grained features from the early backbone layers. By concatenating shallow feature maps with the outputs of the SPPF module, the fusion combines multiscale contextual information with fine-grained details. The CMSFF-YOLOv8 neck is a bridge between the detection head and the backbone to detect objects of different sizes, P3–P5. The CMSFF-YOLOv8 head is predicting bounding boxes and

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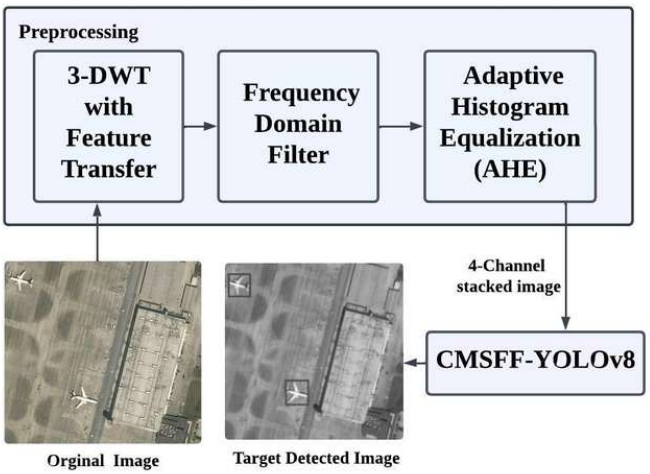


Fig. 1. Overall architecture diagram of the proposed model.

class probabilities. We have enhanced this component by integrating FFM, which improves accuracy in detecting objects across various scales, with a particular focus on small object detection.

The main adjustments of CMSFF-YOLOv8 include a custom backbone that better isolates the required features of aircraft from images taken from a satellite view, leading to improved feature extraction. The Unit Box also has operational capacity, allowing it to focus on crucial areas of the image so that the detection rate can be more adaptive to challenges with accurate and reliable results in the most difficult backgrounds. Multiscale features from preprocessing are enhanced by SEConv and MGEFA. These features are combined using MSFF. These features help to capture different sizes of aircraft effectively. Such improvements solve the difficulties associated with satellite imagery, including diverse crafts, orientations, and complicated backgrounds.

III. Method

The proposed method integrates the CMSFF-YOLOv8 architecture with enhanced preprocessing approaches, including multilevel DWT, wavelet FT, FDF, and AHE, to form a four-channel stacked image representation. The overall architecture diagram of the proposed model is shown in Fig. 1.

A. Preprocessing

If $W_\psi(a,b)$ represents the wavelet coefficients at scale a and translation b, $I(m,n)$ is the input image or signal, $\psi_{a,b}(m,n)$ is the wavelet function (mother wavelet) scaled by a and translated by b, and the summation runs over all the pixel coordinates (m,n), then the general equation of the DWT can be written as

$$W_\psi(a,b) = \sum_m \sum_n I(m,n) \cdot \psi_{a,b}(m,n). \tag{1}$$

We have updated the general equation of the DWT to incorporate FT terms according to our method. Here are the modified equations for each subband at the decomposition level ($i = 1,2,3$). If the original image is represented by $\mathrm{LL}_0(a,b) = I(a,b)$, where the low-pass filter and high-pass filter are represented by e and f, respectively, and the FT scaling factor is represented by α, then we can write the equations as follows.

Approximation Subband (LL_i): It computes the approximation subband (LL_i), capturing coarse, low-frequency information without FT to prevent accumulated noise

$$\mathrm{LL}_i(a,b) = \sum_{m,n} \mathrm{LL}_{i-1}(m,n) \cdot e(a-m) \cdot e(b-n). \tag{2}$$

Horizontal Detail Subband (LH_i): It computes the horizontal detail subband (LH_i), preserving horizontal edge features and textures, with recursive FT from the previous level

$$\mathrm{LH}_i(a,b) = \sum_{m,n} \mathrm{LL}_{i-1}(m,n) \cdot e(a-m) \cdot f(b-n) + \alpha \cdot \mathrm{LH}_{i-1}(a,b). \tag{3}$$

Vertical Detail Subband (HL_i): It computes the vertical detail subband (HL_i), highlighting vertical structures and maintaining continuity across levels using scaled transfer

$$\mathrm{HL}_i(a,b) = \sum_{m,n} \mathrm{LL}_{i-1}(m,n) \cdot f(a-m) \cdot e(b-n) + \alpha \cdot \mathrm{HL}_{i-1}(a,b). \tag{4}$$

Diagonal Detail Subband (HH_i): It computes the diagonal detail subband (HH_i), enhancing corner and diagonal texture features through the recursive integration of prior-level details

$$\mathrm{HH}_i(a,b) = \sum_{m,n} \mathrm{LL}_{i-1}(m,n) \cdot e(a-m) \cdot f(b-n) + \alpha \cdot \mathrm{HH}_{i-1}(a,b) \tag{5}$$

where α is the scaling factor that controls the FT between decomposition levels. Selecting the appropriate α results in the retention and dissemination of important features (such as edges) across different levels. Although higher α values may increase noise, they can also improve the identification of small objects. Depending on the nature of our application, a moderate range of $\alpha = 0.5$–1.0 would be optimal for aircraft identification in satellite images. In fact, this effect is visually noticeable in the output images. For further validation, we determined that the best performance is achieved from the experiment with a scaling factor $\alpha = 0.75$. We used a scaling factor $\alpha = 0.75$ to balance feature preservation and noise reduction in the model. These formulas show the DWT's recursive structure, in which each level further breaks down the approximation subband of the previous level, while the detail subbands take FT from the corresponding subbands of the level before it.

Fig. 2 shows the detailed architectural diagram with the multilevel DWT; it can break down images into frequency subbands like LL (low-frequency component) and HL, LH, and HH (high-frequency coefficient components). Different levels of detail are captured by these subbands. A method for improving image contrast and illumination that works on global histogram equalization and is not suitable for increasing local details of the image is called AHE [36], [38], [39]. AHE allows for localized contrast improvement by working on specific parts of the image, as opposed to standard histogram

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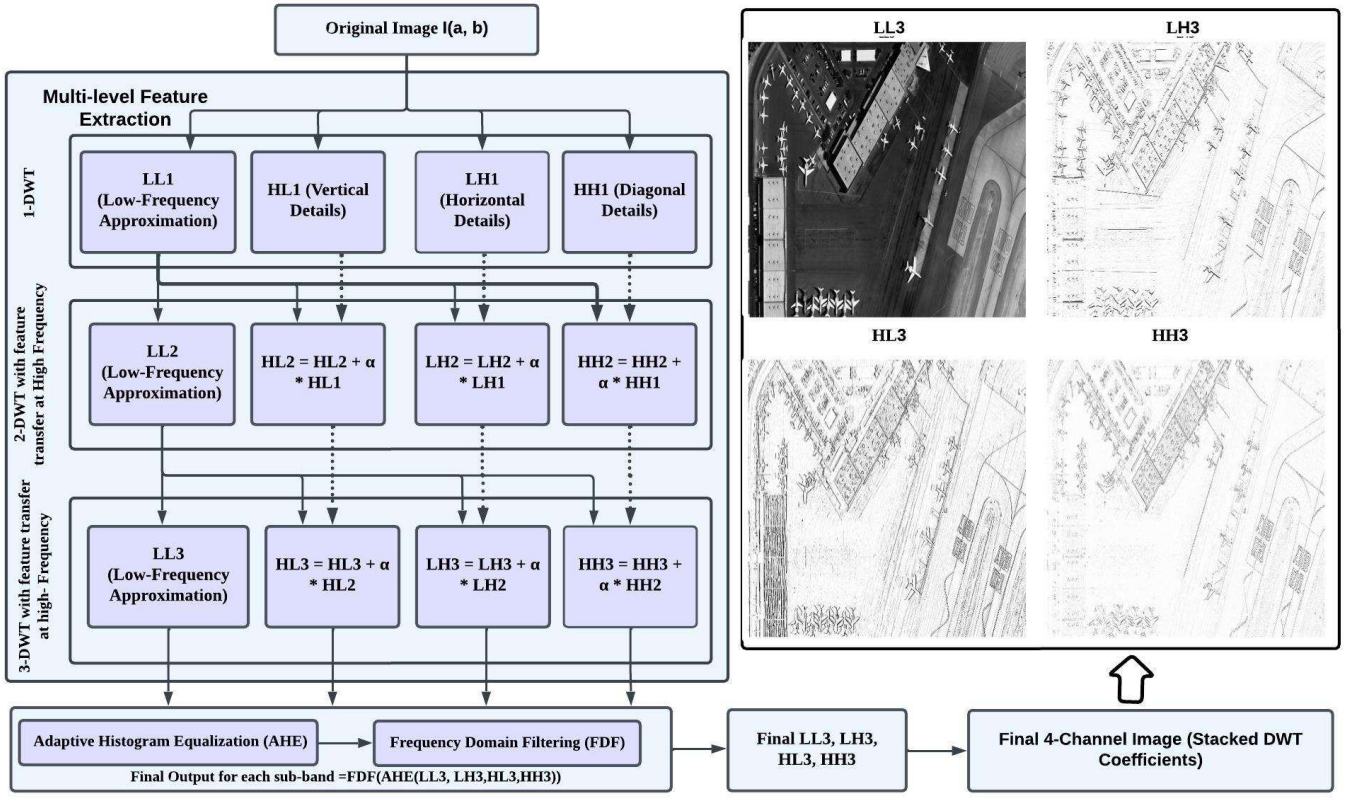


Fig. 2. Detailed architectural diagram of the preprocessing step of the proposed method.

equalization, which applies a uniform alteration derived from the global histogram. Fig. 3 illustrates the analysis of all preprocessing outputs, including DWT, AHE, FT, and FDF.

The high-frequency subbands (HL_3, LH_3, HH_3) coefficients are subjected to AHE to improve local contrast and edge details. The procedure involves the following.

1) *Divide the Subband Into Tiles:* Each subband is partitioned into nonoverlapping tiles.
2) *Compute Local Histograms:* For each tile, calculate the histogram representing the distribution of coefficient magnitudes.
3) *Calculate the Cumulative Distribution Function (cdf):* For each tile, compute the cdf from the local histogram.

This method effectively enhances local contrast and edge details by focusing on the high-frequency components at the third level of DWT decomposition while preserving the overall image structure represented by the LL_3 subband. Using these third-level decomposition coefficients, its four-channel stacked images are constructed. These four-channel stacked images are then trained using a CMSFF-YOLOv8 architecture. This approach aligns with the principles of multilevel feature extraction, where hierarchical features are analyzed to improve object detection performance.

Map Coefficients Using the CDF: Each coefficient c on the tile is mapped to a new value c' using the cdf. If the minimum nonzero value of the cdf in the tile is $\text{cdf}_{\min}$ and the total number of coefficients in the tile is $M \times N$, then

$$c' = \text{round}\left(\frac{\text{cdf}(c) - \text{cdf}_{\min}}{(M \times N) - \text{cdf}_{\min}} \times (L-1)\right). \tag{6}$$

Let AHE $(\cdot)$ denote the AHE function. The enhanced subbands are

$$\text{LH3}' = \text{AHE}(\text{LH3}) \tag{7}$$

$$\text{HL3}' = \text{AHE}(\text{HL3}) \tag{8}$$

$$\text{HH3}' = \text{AHE}(\text{HH3}). \tag{9}$$

This method effectively enhances local contrast and edge details by focusing on the high-frequency components at the third level of the DWT decomposition while preserving the overall image structure represented by the LL_3 subbands. Using these third-level decomposition coefficients, its four-channel stacked images are constructed. These four-channel stacked images are then trained using a CMSFF-YOLOv8 architecture. This approach aligns with the principles of multilevel feature extraction, where hierarchical features are analyzed to improve object detection performance.

To suppress specific frequency components, FDF is applied to the high-frequency subbands (HL, LH, and HH) of the DWT coefficients. We introduce FDF() as a function that applies frequency-based filtering (e.g., high pass) to the high-frequency components after AHE

$$F'_{\text{HF}}(u,v) = H_{\text{FDF}}(u,v) \cdot F_{\text{HF}}(u,v) \tag{10}$$

where $H_{\text{FDF}}(u,v)$ denotes the frequency-domain filter applied to the high-frequency components $F_{\text{HF}}(u,v)$. This filter typically functions as a high-pass filter, enhancing fine details and textures. The variables (u,v) represent the horizontal and vertical frequency coordinates. The resulting component, $F'_{\text{HF}}(u,v)$, corresponds to the filtered high-frequency content

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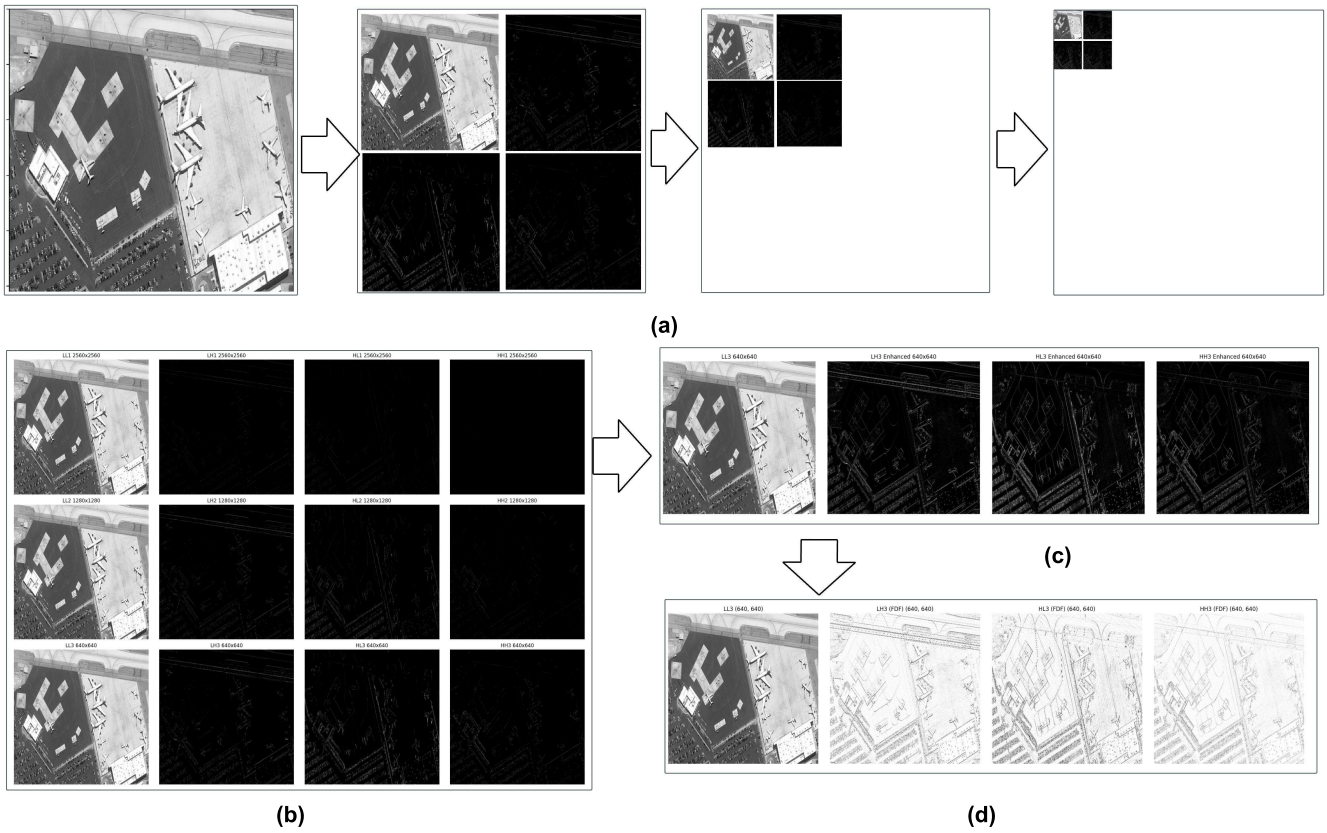


Fig. 3. Detailed analysis of preprocessing output image for DWT decomposition, FT, AHE, and FDF. (a) Analysis of the original image and its three levels of decomposition result without enhancements. (b) Enhanced FT for DWT. (c) Enhanced AHE at third-level decomposition. (d) Enhanced FDF at third-level decomposition.

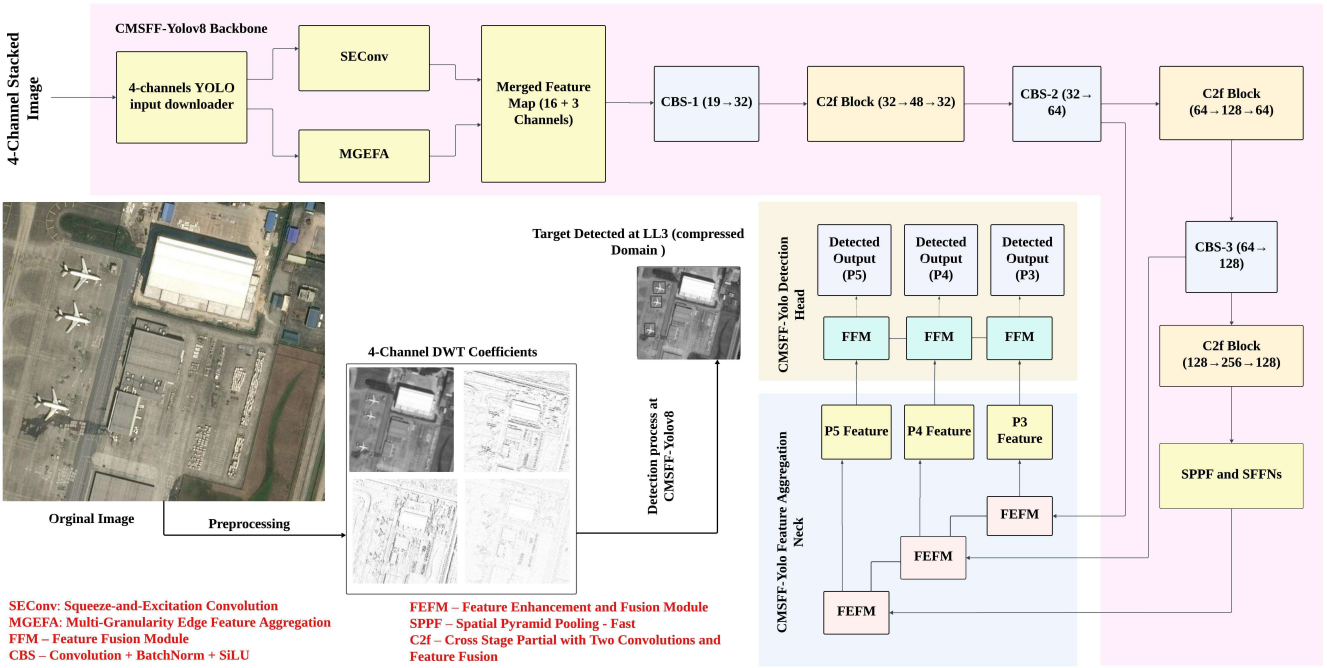


Fig. 4. Detailed architectural diagram of the proposed CMSFF-YOLOv8.

in the frequency domain, capturing the enhanced or modified version of $F_{\mathrm{HF}}(u, v)$ after filtering. The image is reconstructed using the inverse DWT (IDWT), which blends the enhanced detail subbands with the initial approximation subband after

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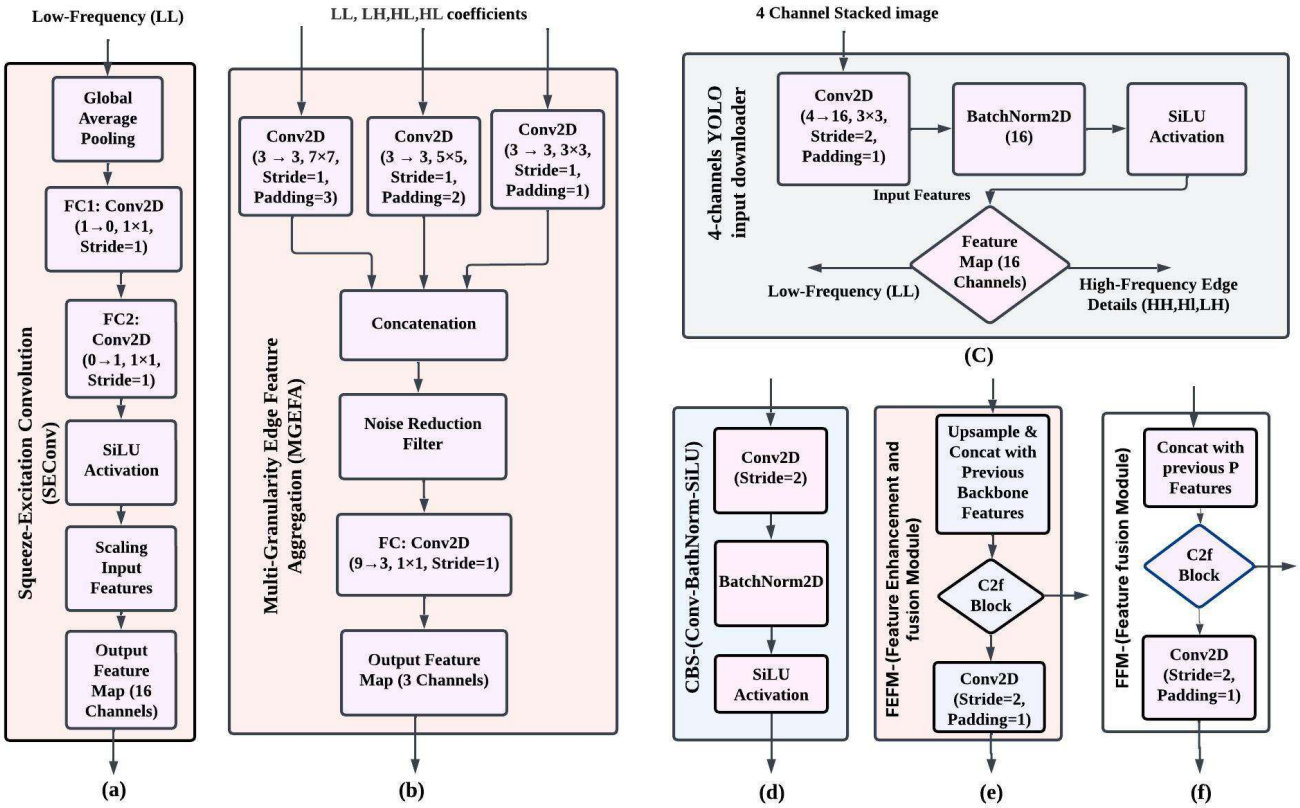


Fig. 5. Detailed components of the CMSFF-YOLOv8 architecture. (a) SEConv module. (b) MGEFA module. (c) Input layer module. (d) CBS module. (e) FEFM module. (f) FFM module.

the high-frequency subbands of the DWT have been enhanced

$$(\mathrm{LL}_2): I'(a,b) = \mathrm{IDWT}(\mathrm{LL}_3, \mathrm{FDF}(\mathrm{AHE}(\mathrm{LH3}' + \alpha\mathrm{LH2}')), \mathrm{FDF}(\mathrm{AHE}(\mathrm{HL3}' + \alpha\mathrm{HL2}')) \mathrm{FDF}(\mathrm{AHE}(\mathrm{HH3}' + \alpha\mathrm{HH2}'))) \quad (11)$$

$$(\mathrm{LL}_1): I'(a,b) = \mathrm{IDWT}(\mathrm{LL}_2, \mathrm{FDF}(\mathrm{AHE}(\mathrm{LH2}' + \alpha\mathrm{LH1}')), \mathrm{FDF}(\mathrm{AHE}(\mathrm{HL2}' + \alpha\mathrm{HL1}')) \mathrm{FDF}(\mathrm{AHE}(\mathrm{HH2}' + \alpha\mathrm{HH1}'))) \quad (12)$$

$$(\mathrm{LL}_0): I'(a,b) = \mathrm{IDWT}(\mathrm{LL}_1, \mathrm{FDF}(\mathrm{LH1}') \mathrm{FDF}(\mathrm{HL1}'), \mathrm{FDF}(\mathrm{HH1}')). \quad (13)$$

Equations (11)–(13) demonstrate that after applying the DWT, FT, AHE, and FDF modules, the original image can be effectively reconstructed, confirming the reversibility of the proposed enhancement process as well as the overall workflow, thereby ensuring that no critical spatial information is lost.

B. CMSFF-YOLOv8 Backbone Stage

The detailed architectural diagram of the proposed CMSFF-YOLOv8, as shown in Fig. 4, introduces several modifications to the standard YOLOv8 model to accommodate four-channel input images and enhance feature extraction. Both the CMSFF-YOLOv8 architecture and YOLOv8 models for comparison of their performance with our modified model contribute significantly to the advancement of object detection models. The CMSFF-YOLOv8's integration of modules like MGEFA and enhanced SPPF layers addresses specific challenges in complex detection scenarios, while YOLOv8's architectural refinements and training strategies offer a robust, general-purpose solution with improved performance metrics. These developments, which meet a range of application requirements and increase the capacity for real-time object identification, are a reflection of the field's continuous evolution and specialization.

The essential elements of the CMSFF-YOLOv8 architecture are represented mathematically as follows.

1) Input Layer: The input layer or modified four-channel input downloader is adapted to accept four-channel images resulting from our preprocessing steps. This is achieved by modifying the first convolutional layer to handle an input with four channels instead of the standard three. The detailed modification of the input modules at the backbone of the CMSFF-YOLOv8 architecture is shown in Fig. 5(c). Let $I \in \mathbb{R}^{H\times W\times 4}$ represent the input image with four channels, height (H), and width (W); $W_1 \in \mathbb{R}^{16\times 4\times 3\times 3}$ represent the weight tensor of the first convolutional layer, which has four input channels and 16 3 $\times$ 3 filters; and $b_1 \in \mathbb{R}^{16}$ be the bias vector for this layer. The first convolutional layer's output (Y_1) is computed as

$$Y_1 = \mathrm{SiLU}\left(\mathrm{BN}\left(\mathrm{Conv}\left(I, W_1\right) + b_1\right)\right) \quad (14)$$

where: 1) $\mathrm{Conv}(I, W_1)$ indicates the process of convolution between the weight tensor W_1 and the input image I; 2) BN(·)

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denotes batch normalization; and 3) SiLU(·) is the activation function applied element-wise.

2) Modified Backbone With SEConv and MGEFA Modules: The backbone incorporates SEConv and multigranularity edge feature aggregation modules to enhance feature representation, improve the model's ability to capture dependencies, and make the network more sensitive to important features. The detailed modification of the SEConv and MGEFA modules at the backbone of the CMSFF-YOLOv8 architecture is shown in Fig. 5(a) and (b).

SEConv Module: It applies channel-wise attention to recalibrate feature maps. The input feature map with C channels is represented by $X \in \mathbb{R}^{C\times H\times W}$, and the output feature map following SEConv and the SEConv operation is indicated by $F \in \mathbb{R}^{C\times H\times W}$. It is defined as

$$F = X \otimes \sigma\left(\mathrm{SE}\left(X\right)\right) \tag{15}$$

where: 1) $\otimes$ denotes that the channel-wise weights are being broadcast across spatial dimensions; 2) $\sigma(\cdot)$ is the sigmoid activation function; and 3) **SE**(X) is defined as follows:

$$\mathbf{SE}\left(X\right) = \mathrm{FC}_2\left(\mathrm{ReLU}\left(\mathrm{FC}_1\left(\mathrm{GAP}\left(X\right)\right)\right)\right) \tag{16}$$

performs element-wise multiplication, and GAP(X) computes the global average pooling of X, resulting in a vector $z \in \mathbb{R}^C$ with components

$$z_c = \frac{1}{H\times W}\sum_{i=1}^{H}\sum_{j=1}^{W} X_{c,i,j}. \tag{17}$$

FC1 and FC2 are fully connected layers that reduce and then restore the channel dimensionality, respectively. If $W_1 \in \mathbb{R}^{C\times r}$ and $W_2 \in \mathbb{R}^{C\times r}$ are under the reduction ratio r, and ReLU(·) is the rectified linear unit, then we have

$$\mathrm{FC}_1\left(z\right) = W_1 z + b_1 \quad \text{and} \quad \mathrm{FC}_2\left(z\right) = W_2 z + b_2. \tag{18}$$

MGEFA Module: It captures multiscale edge features using convolutions of varying kernel sizes. Let $X \in \mathbb{R}^{C\times H\times W}$ be the input feature map F_3, F_5, and F_7 representing the feature maps obtained using 3×3, 5×5, and 7×7 convolutions, respectively. $F \in \mathbb{R}^{C\times H\times W}$ denotes the aggregated feature map after MGEFA. Then, the MGEFA operation is defined as

$$F = \mathrm{Conv}_{1\times 1}\left(\left[F_3; F_5; F_7\right]\right) \tag{19}$$

where: 1) $F_3 = \mathrm{Conv}_3(X)$, $F_5 = \mathrm{Conv}_5(X)$, and $F_7 = \mathrm{Conv}_7(X)$; 2) $[F_3; F_5; F_7]$ denotes the concatenation of the feature maps along the channel dimension, producing a tensor of shape $\mathbb{R}^{3C\times H\times W}$; and 3) $\mathrm{Conv}_{1\times 1}(\cdot)$ applies a 1×1 convolution that fuses the concatenated features and reduces the channel dimensions to C once again.

After applying the SEConv and MGEFA modules, the features are combined at multiple scales through feature fusion and feature enhancement. This aggregation is crucial for making the model aware of both high-level context and fine-grained details.

3) Merging the Features: After processing the LL coefficients and DWT-derived features through the SEConv and MGEFA modules, respectively, we merge the feature maps

$$\begin{aligned}&\text{Merged Feature Map}\\ &= \big[\text{SEConv Feature Map (16 channels)};\\ &\qquad \text{MGEFA Feature Map (3 channels)}\big].\end{aligned} \tag{20}$$

This results in a merged feature map with 19 channels. The merged feature map now includes both the enhanced LL features and the multiscale edge features from the DWT coefficients.

4) Conv-BatchNorm-SiLU: This block is a basic building block commonly used in modern neural networks. It consists of three components: convolution, batch normalization (BN), and sigmoid linear unit (SiLU) activation. The detailed modification of the Conv-BatchNorm-SiLU (CBS) modules at the backbone of the CMSFF-YOLOv8 architecture is shown in Fig. 5(d). For convolution (Conv), the convolution layer applies a convolutional filter over the input image to extract features. The convolution operation can be defined as

$$\mathrm{Output}\left(a,b\right) = \sum_{m,n}\mathrm{Input}\left(m,n\right)\cdot W\left(a-m,b-n\right) + b \tag{21}$$

where: 1) Input(m,n) is the input feature map (the image or the feature map from the previous layer); 2) $W(a-m,b-n)$ is the filter applied to the input image; and 3) b is the bias term.

BN: After convolution, BN is applied to stabilize and accelerate training by normalizing the output of each mini-batch. The equation for BN is

$$\hat{x} = \frac{x-\mu}{\sqrt{\sigma^2+\epsilon}} \tag{22}$$

$$y = \gamma\hat{x} + \beta \tag{23}$$

where: 1) x is the input to the BN layer; 2) μ and σ^2 are the mean and variance of the mini-batch, respectively; 3) γ and β are learnable scaling and shifting parameters, respectively; and 4) ϵ is a small constant for numerical stability.

SiLU Activation: The SiLU activation function, also known as the swish function, is applied element-wise

$$\mathrm{SiLU}\left(x\right) = x\cdot\sigma\left(x\right) \tag{24}$$

where $\sigma(x) = \frac{1}{1+\exp(-x)}$ is the sigmoid function.

5) SPPF: SPPF is a module used in the modified YOLOv8 architecture, aimed at enhancing the model's ability to process images at different scales, improving its robustness for various object detection tasks. Given an input feature map $X \in \mathbb{R}^{C\times H\times W}$, the module applies convolutions of varying kernel sizes (such as 1×1, 3×3, and 5×5) to extract features at different scales. SFFNs enhance detection by integrating fine-grained features from the early layers. When we combine SFFNs with SPPF, this approach effectively captures both detailed spatial information and multiscale contextual features. By concatenating shallow feature maps with the outputs of the SPPF module, the fusion combines multiscale contextual information with fine-grained details, thereby improving detection accuracy for tiny objects.

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C. CMSFF-YOLOv8 Feature Aggregation Neck

These are the feature maps extracted at different stages of the CMSFF-YOLOv8 backbone. They correspond to different levels of abstraction, which capture both fine-grained details (small objects) and high-level information (large objects). P3 comes from a shallow layer, P4 comes from a middle layer, and P5 comes from a deeper layer in the network. It aggregates multiscale features through upsampling, concatenation, and convolution, resulting in a combined feature map that highlights information across multiple scales. The feature enhancement and fusion module (FEFM) is responsible for combining features from different sources to enhance the learned features and improve the model's representation capabilities. The detailed modification of the FEFM modules at the backbone of the CMSFF-YOLOv8 architecture is shown in Fig. 5(e). For upsampling concatenation, the FEFM begins by upsampling and concatenating features from the previous backbone layers (such as P3 features). The upsampling process increases the resolution of the feature maps, making them more suitable for detection at different scales. Upsampling is done via

$$\text{Upsample}(X) = \text{Interpolated}(X, \text{factor}).$$

The concatenation of feature maps from different levels is

$$\text{Concat}(X, Y) = \text{Concat}(X, Y, \text{axis}).$$

Cross-Stage Partial With Two Convolutional Layers and Feature Fusion (C2f) Block: It, which is a combination of multiple convolutional operations, helps to fuse different scales of feature maps. The operation is given by

$$\text{C2f}(X) = \text{Conv2D}(\text{Conv1}(X), \text{Conv2}(\text{Conv1}(X))) \quad (25)$$

where Conv1(X) and Conv2(X) are convolutions applied to the input X.

Thus, FEFM provides a more discriminative feature representation by enhancing the features from the backbone and fusing them across different scales.

D. CMSFF-YOLOv8 Detection Head

The CMSFF-YOLOv8 detection head is responsible for generating the final detection outputs, which include bounding boxes, object classes, and confidence scores. It operates on the aggregated features produced by the feature aggregation neck. Each of the detection outputs in P_3–P_5 is processed to provide the following.

1) *Class Predictions:* $C \in \mathbb{R}^{N_{\text{classes}}}$, representing the class probabilities for each detected object.
2) *Bounding Box Coordinates:* $B = (x, y, w, h)$, representing the center coordinates and the width/height of the bounding box.
3) *Confidence Score:* $S \in [0, 1]$, representing the model's confidence that the detected object belongs to the predicted class.

FFM: It is responsible for fusing multiple feature maps obtained from different levels (such as P3–P5 features) to improve detection accuracy, especially for objects at different scales. The detailed modification of the FFM modules at the backbone of the CMSFF-YOLOv8 architecture is shown in Fig. 5(f). For concatenation, similar to the FEFM, FFM concatenates features from various levels of the network

$$\text{Concat}(P_3, P_4, P_5) = \text{Concat}(P_3, P_4, P_5, \text{axis})$$

where P_3–P_5 are feature maps from different levels of the network. For C2f block, after concatenating the features, the C2f block is applied to refine and merge the features. As explained before

$$\text{C2f}(X) = \text{Conv2D}(\text{Conv1}(X), \text{Conv2}(\text{Conv1}(X))). \quad (26)$$

Convolution: Finally, a convolution is applied to the concatenated and merged features to further process the feature map before passing it to the next layers

$$\text{Conv2D}(X).$$

Thus, FFM plays an important role in aggregating information from different layers to enhance the detection capabilities of the model.

IV. Experimental Results

We scale images to a fixed size of 5120 × 5120 pixels to ensure consistent processing of both images and their annotations, making it easier to test different datasets with varying sizes. The aspect ratio remains the same after the resizing process. The YOLO-formatted annotation file for every image is adjusted to fit the updated image dimensions. This file includes the normalized class ID, center coordinates, width, and height. After performing the 3-DWT decomposition, the final output of preprocessing consists of a four-channel image, where each channel represents one of the subbands of the decomposition (LL3, LH3, HL3, and HH3). After 3-DWT, the image size is reduced to 640 × 640 pixels, ensuring that all channels have the same dimensions. Each subband is normalized to the 8-bit range (0–255), and the resulting four-channel image is saved in TIFF format for high-quality storage.

A. Datasets

We conducted experiments on our experimental evaluation by incorporating multiple publicly available datasets, namely, HRPlanesv2, RarePlanes, and instance segmentation in aerial imagery (iSAID). First, the HRPlanesv2 dataset [40] includes 2120 Google Earth high-resolution photographs, labeled with 14 335 aircraft across various regions, including civilian, military, and joint-use airports. The images are in ".jpg" format, and the YOLOv8 ".txt" format is used for the corresponding annotations. Second, RarePlanes [41] is an open-source machine learning dataset developed by CosmiQ Works and AI. Reverie, which includes both real and synthetic satellite imagery, has the purpose of assessing the potential of synthetic data to enhance aircraft categorization and detection. Approximately 630 000 aircraft annotations and 50 000 simulated photographs are produced by AI. Reverie's simulation platform is included in the synthetic dataset.

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Third, the iSAID dataset [42] is a large-scale benchmark designed for instance segmentation in remote sensing. It contains 655 451 object instances across 15 categories, including airplanes, vehicles, and ships. Each image is of high resolution (800×800 pixels or more), and annotations are provided in the polygon format, which we converted to bounding boxes for use in YOLOv8. Additionally, the High Resolution Planes dataset [43] provides 3092 RGB images from major airports and aircraft boneyards, with 18 477 airplanes labeled using bounding boxes. In addition to evaluating individual datasets, we also created a mixed dataset by combining representative samples from the HRPlanesV2 and RarePlanes datasets to enhance diversity in aircraft types, imaging conditions, and background complexity. Training uses 70% of the datasets, validation uses 20% of the datasets, and testing uses the remaining portion of all datasets.

B. Implementation Details

The proposed preprocessing model, applied across all datasets, utilizes a complete image processing pipeline that incorporates AHE, FDF, and DWT to enhance features in satellite images. Before undergoing multilevel DWT decomposition to extract frequency components at different scales, the images are preprocessed to increase contrast and smooth the edge details for small object detection. FDF blocks or reduces low-frequency components in high-frequency coefficients to highlight edges and finer details, enhancing feature extraction and edge augmentation by suppressing smoother areas of the image. The AHE enhances contrast in an attempt to facilitate the discrimination between large planes and smaller planes. As a result, subsequent detection approaches will be improved, as shown in Fig. 3. This setting helps both with plane recognition in satellite images and multiscale feature extraction in the CMSFF-YOLOv8 architecture. The proposed CMSFF-YOLOv8 architecture introduces several modifications to the standard YOLOv8 model, including the integration of the SFFNs module with the SPPF layers, enhancing the MGEFA, SEConv, FFM, FEFM, and so on. These developments address a variety of application requirements together push the limits of real-time object detection.

We conducted a series of experiments on the HRPlanesv2, RarePlanes, and iSAID datasets, exploring different aspects of aircraft detection. In the first model (Model 1), we applied a 3-DWT to convert the image into a four-channel format without preprocessing enhancements and then trained the standard YOLOv8 model with modifications to support a four-channel input loader. In the second model (Model 2), we applied a 3-DWT to convert the image into a four-channel format with preprocessing enhancements. We then trained the standard YOLOv8 model with modifications to support a four-channel input loader, resulting in enhanced preprocessing performance of our proposed model. In the third model (Model 3), we applied a 3-DWT to convert the image into a four-channel format with preprocessing enhancements and then trained only the LL3 channel of preprocessing on the standard YOLOv8 model. It provides performance when we train only the LL3 coefficient of the 3-DWT with the standard YOLOv8 model. In the final model (Model 4), we applied a 3-DWT to convert the image into a four-channel format with preprocessing enhancements and then trained the proposed CMSFF-YOLOv8 model with outstanding performance. Model 4 analyzes the contribution of the fusion of FEFM and FFM in our work, while Model 5 evaluates the contribution of the fusion of the SFFNs module with SPPF.

C. Experimental Setup

The proposed CMSFF-YOLOv8 model was trained and evaluated in a Python 3.10.15 environment using PyTorch 2.5.1+cu124. The hardware setup included an NVIDIA GeForce GTX 1080 Ti GPU with 11 172 MiB of memory, as verified by the nvidia-smi tool output. The model achieved an inference speed of 82.02 frames/s (FPS), with a parameter count of approximately 3.01 million and a storage size of 6.00 MB. Although it runs marginally slower than the baseline YOLOv8 uncompressed model, it is faster than the compressed model. CMSFF-YOLOv8 delivers improved detection accuracy by integrating DWT preprocessing and FT modules. These enhancements are specifically optimized for batch-level parallelism, enabling real-time performance while preserving critical frequency-domain features.

The model's architecture was adjusted to detect aircraft in satellite images. This training setting aimed to optimize the model for the task of real-time plane detection. The model was trained for 200 epochs using the same batch size of 2; the image size was 640×640 pixels, and the AdamW optimizer with an initial learning rate of 0.002, a momentum of 0.9, and a weight decay of 0.0005 was employed to facilitate convergence and generalization. Automatic mixed precision (AMP) was turned on to improve training speed. The training dataset contained only one class (plane), with the configuration specification indicating one class. More robust data augmentation techniques were implemented to enhance the model's robustness and diversify the training data. The procedure included some mosaic augmentation (i.e., combining four images together) and adjustment of color hue, saturation, brightness, and so on. Additionally, we applied data augmentation techniques such as random flipping, rotation, scaling, translation, and shearing that helped account for different orientations and human perspectives of the objects. To increase diversity in the results, augmentations such as perspective distortion, blur, random noise, and zoom were manually added, which were beneficial for enabling the model to train more effectively in realistic situations.

D. Calculations

1) Evaluation Metrics: We load performance metrics of the training and validation experiments from CSV files generated by the model that records output. Some of the key evaluation metrics include mean average precision (mAP) at intersection over union (IoU) thresholds, such as mAP at IoU = 0.5 (mAP@0.5) and mAP at IoU = 0.75 (mAP@0.75). There can be fewer than a dozen numbers (75) in these files. Precision and recall values are also extracted to measure the model's performance in terms of completeness and its accuracy in predicting objects. These extracted data are then stored in a dictionary called metrics, where it is associated with its corresponding metric and an array representing the metrics of each model (Model 1, Model 2, and so on).

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TABLE I
COMPARISON OF PERFORMANCE METRICS WITH DIFFERENT BASELINE MODELS AND DATASETS

Model	mAP@0.5	mAP@0.75	Precision	Recall	F1-Score
YOLOv8 - HRPlanesv2 datasets (Compressed Mode)	0.704	0.4038	0.907 (AP)	0.726	0.806
FFCA-YOLO with VEDAI, AI-TOD datasets (Uncompressed Mode)	0.909	–	0.929	0.855	0.858
L-FFCA-YOLO with VEDAI, AI-TOD datasets (Uncompressed Mode)	0.909	–	0.929	0.851	0.845
CMSFF-YOLOv8 - HRPlanesv2 datasets (Compressed Mode)	**0.90895**	**0.60063**	**0.87460**	**0.82165**	**0.8473**
CMSFF-YOLOv8 - Rareplanes datasets (Compressed Mode)	0.90253	0.55840	0.86634	0.82931	0.8421
CMSFF-YOLOv8 - iSAID datasets (Compressed Mode)	0.90535	0.54092	0.87064	0.82642	0.8536
CMSFF-YOLOv8 – Mixed datasets (RarePlanes, iSAID, HRPlanes, Compressed Mode)	**0.90155**	**0.63496**	**0.89064**	**0.83632**	**0.8626**

2) Mean Average Precision at IoU Threshold: The average precision is at a threshold of 0.5 for IoU. This evaluates the model's accuracy by calculating the average precision for all classes at IoU = 0.5

$$\text{mAP@0.5} = \frac{1}{N}\sum_{i=1}^{N} \text{AP}_i \quad \text{for IoU=0.5} \tag{27}$$

where N is the number of classes and AP_i is the average precision for class i.

For calculating the average precision for all classes at IoU = 0.75, the formula remains the same as for IoU = 0.5, but it is applied at IoU = 0.75.

3) Precision: Precision derives from how precise the model is in predicting good results or how many of the expected objects were correctly predicted as positive predictions. The formula for precision is

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{28}$$

where TP is the number of true positives (correctly identified objects) and FP is the number of false positives (incorrectly identified objects).

4) Recall: Recall measures the completeness of the positive predictions, i.e., how many actual positive objects were correctly examined by the model. The formula for recall is

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{29}$$

where TP is the number of true positives (correctly identified objects) and FN is the number of false negatives (objects that were missed by the model).

5) F1-Score Calculation: The $F1$-score is a crucial performance measure that combines precision and recall into a single value, which is especially useful when there is an imbalance in the dataset. It is calculated using the formula

$$F1 = \frac{2 \cdot (\text{Precision} \cdot \text{Recall})}{\text{Precision} + \text{Recall}}. \tag{30}$$

This formula guaranties that false positives (false catches) and false negatives (missed detections) weigh equally against each other, which is convenient when our data are not balanced. Next, the calculated $F1$-scores for each model are also appended to the metrics dictionary, which contains values for mAP, precision, and recall.

6) DataFrame Creation: After pulling everything needed and calculating $F1$-scores, we have to format these data into a readable format. This is achieved by constructing a pandas DataFrame, a 2-D data structure that makes data manipulation at the table level simple and easy to present. The metrics dictionary, which contains all the extracted metrics and calculated $F1$-scores, is used to create the DataFrame. The model names (Model 1–Model 4) serve as the index (row labels) of the DataFrame, while the metrics (mAP@0.5, mAP@0.75, precision, recall, and $F1$-Score) are displayed as columns. This DataFrame provides a clear, tabular representation of the performance comparison across the different models, making it easier to analyze and compare the results.

E. Result Comparison

To evaluate the performance, we carried out compressed-domain aircraft detection experiments with YOLOv8 using the HRPlanesv2, RarePlanes, and iSAID datasets. Without compression, various algorithms [44] demonstrate differing levels of performance, as shown in Table I. The proposed baseline architecture is evaluated on remote sensing datasets such as VEDAI and AI-TOD [8] using techniques like feature fusion and channel attention YOLO (FFCA-YOLO) and lightweight feature fusion and channel attention YOLO (L-FFCA-YOLO), which improve YOLOv5 for small object detection through feature enhancement, fusion, and spatial context modules without incorporating compression. In contrast, our approach integrates frequency-domain information via 3-DWT compression. FFCA-YOLO reports mAP@0.5 scores of 0.748 on VEDAI, 0.617 on AI-TOD, and 0.909 on the private USOD dataset. Our CMSFF-YOLOv8 achieves a comparable 0.90895 mAP@0.5 on HRPlanesv2, with even better results on RarePlanes and iSAID. This is also evaluated and presented in Table II, highlighting the performance advantages of our method.

The extended sensitivity analysis of the α (alpha) parameter for three datasets—HRPlanesV2, RarePlanes, and iSAID—is presented in Table III, which evaluates the effect of varying α on detection and classification performance. For each dataset, performance metrics, including precision, recall, $F1$-score, mAP@0.50, and mAP@0.75, are reported across α values ranging from 0.5 to 1.0. The results show that moderate α values (particularly around 0.75–0.8) yield the most balanced performance across all datasets, achieving relatively higher $F1$-scores ($\approx$ 0.85–0.87) and mAP values. In contrast, lower or higher α values lead to a gradual decrease in both precision

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TABLE II
PERFORMANCE METRICS FOR PREPROCESSING METHODS IN YOLOV8 BASELINE MODELS AND THE CMSFF-YOLOV8 MODEL

Model	mAP@0.5	mAP@0.75	Precision	Recall	F1-Score
HRPlanesv2 datasets-Model 1 (YOLOv8 with 3-DWT with single channel image)	0.84965	0.55840	0.83358	0.75868	0.794368
Rareplanes datasets- Model 1 (YOLOv8 with 3-DWT with single channel image)	0.60436	0.37065	0.75815	0.60903	0.67545
iSAID datasets- Model 1 (YOLOv8 with 3-DWT with one channel image)	0.68184	0.43643	0.75815	0.60903	0.67545
HRPlanesv2 datasets- Model 2 (YOLOv8 with enhanced 4- channel 3-DWT)	0.84965	0.55840	0.83538	0.75688	0.79636
Rareplanes datasets- Model 2 (YOLOv8 with enhanced 4- channel 3-DWT)	0.83358	0.54942	0.81900	0.75650	0.73580
iSAID datasets- Model 2 (YOLOv8 with enhanced 4- channel 3-DWT)	0.84368	0.54092	0.82038	0.77035	0.82169
HRPlanesv2 datasets- Model 3 (YOLOv8 with enhanced LL3 of 3-DWT)	0.88368	0.58945	0.88206	0.81083	0.844946
Rareplanes datasets- Model 3 (YOLOv8 with enhanced LL3 of 3-DWT)	0.86380	0.54092	0.88038	0.77035	0.82169
iSAID datasets- Model 3 (YOLOv8 with enhanced LL3 of 3-DWT)	0.87378	0.56087	0.88143	0.79875	0.82583
HRPlanesv2 datasets- Model 4 (CMSFF-YOLOv8 with enhanced 3-DWT)	**0.90895**	**0.60063**	**0.87460**	**0.82165**	**0.847299**
Rareplanes datasets- Model 4 (CMSFF-YOLOv8 with enhanced 3-DWT)	**0.90253**	**0.55840**	**0.86634**	**0.82931**	**0.8421**
iSAID datasets- Model 4 (CMSFF-YOLOv8 with enhanced 3-DWT)	**0.90535**	**0.54092**	**0.87064**	**0.82642**	**0.8536**
HRPlanesv2 datasets- Model 5 (CMSFF-YOLOv8 without Fusion of FEFM and FFM)	0.88157	0.58945	0.85653	0.73654	0.79416
Rareplanes datasets- Model 5 (CMSFF-YOLOv8 without Fusion of FEFM and FFM)	0.87390	0.55952	0.88983	0.77834	0.83129
iSAID datasets- Model 5 (CMSFF-YOLOv8 without Fusion of FEFM and FFM)	0.88318	0.56879	0.88939	0.81825	0.83483
HRPlanesv2 datasets- Model 6 (CMSFF-YOLOv8 without Fusion of SFFNs with SPPF)	0.86845	0.53532	0.84368	0.75218	0.78432
Rareplanes datasets- Model 6 (CMSFF-YOLOv8 without Fusion of SFFNs with SPPF)	0.85941	0.55792	0.83093	0.76741	0.82956
iSAID datasets- Model 6 (CMSFF-YOLOv8 without Fusion of SFFNs with SPPF)	0.86129	0.56849	0.84869	0.74275	0.80473

TABLE III
EXTENDED ALPHA PARAMETER SENSITIVITY ANALYSIS FOR HRPLANESV2, RAREPLANES, AND ISAID DATASETS

Alpha (α)	HRPlanesV2 datasets					RarePlanes datasets					iSAID datasets				
	Precision	Recall	F1	mAP@0.5	mAP@.75	Precision	Recall	F1	mAP@0.5	mAP@.75	Precision	Recall	F1	mAP@0.5	mAP@.75
0.5	0.86	0.80	0.83	0.88	0.65	0.86	0.80	0.83	0.88	0.534	0.86	0.80	0.84	0.88	0.53
0.6	0.863	0.804	0.837	0.883	0.657	0.863	0.804	0.833	0.883	0.5475	0.863	0.804	0.841	0.883	0.533
0.7	0.873	0.802	0.839	0.890	0.673	0.865	0.802	0.837	0.890	0.553	0.867	0.802	0.847	0.890	0.538
0.75	0.874	0.821	0.8472	0.908	0.680	0.866	0.829	0.842	0.9025	0.558	0.870	0.826	0.853	0.905	0.540
0.8	0.870	0.803	0.844	0.900	0.681	0.864	0.803	0.837	0.900	0.551	0.868	0.803	0.844	0.900	0.530
0.9	0.865	0.801	0.844	0.894	0.672	0.860	0.801	0.834	0.894	0.542	0.864	0.801	0.843	0.894	0.528
1.0	0.860	0.800	0.840	0.890	0.667	0.850	0.800	0.830	0.890	0.535	0.840	0.800	0.840	0.890	0.520

and recall, indicating reduced stability in feature weighting. Overall, Table III demonstrates that tuning α effectively influences model sensitivity, with $\alpha \approx 0.75$ providing an optimal tradeoff between detection accuracy and robustness for compressed remote sensing imagery.

Generally, benchmarks without compression exhibit higher accuracy but are huge in size. Without DWT coefficient enhancement, our model performs at an average level in compressed mode, yet still provides advantages due to a reduction in size. However, our proposed model achieves strong results, demonstrating robust object detection capability with good sensitivity. In Table II, the performance of our different models is demonstrated using important metrics including mAP, accuracy, recall, and $F1$-score. Compared to the grayscale DWT-based YOLOv5 model [45], which achieved an $F1$-score of 0.93 on natural scene datasets containing street signs and human faces, our proposed model demonstrates superior performance on challenging satellite datasets. Their method applies grayscale conversion followed by a single-level DWT, retaining only the LL component as input to YOLOv5. Similarly, wavelet transforms are primarily employed for image compression, and object detection performance is evaluated post-decompression using YOLOv5 [46]. While their reported mAP@0.5 is approximately 0.83 on the COCO validation set, the wavelet subbands serve for compression rather than detection, and the model operates on reconstructed RGB images. Our proposed model featuring a four-channel enhanced 3-DWT demonstrates significantly better performance on challenging satellite datasets, achieving an $F1$-score of 0.8473 and mAP@0.5 of 0.90895 on HRPlanesV2. Our approach directly exploits raw frequency-domain information, preserving rich spatial features that are especially beneficial for small object detection in remote sensing applications.

Model 3, which included FDF, AHE, FT, and preprocessing with DWT, then integrated only the LL3 coefficient image with standard YOLOv8, has an mAP of 0.88368 at IoU = 0.5 and 0.58945 at IoU = 0.75. Model 3 outperformed Model 2, which used the same standard YOLOv8 as Model 2 but with preprocessing that introduced the mechanism of integrating FDF, AHE, and FT. Its accuracy, recall, and $F1$-score values were 0.88206, 0.81083, and 0.8449, respectively. Finally, our proposed Model 4, the CMSFF-YOLOv8 model with a four-channel stacked image and sophisticated preprocessing, showed the greatest results, with an $F1$-score of 0.8473, an accuracy of 0.8746, a recall of 0.82165, and an mAP of 0.90895 at IoU = 0.5 and 0.60063 at IoU = 0.75. These findings demonstrate how well multiscale features

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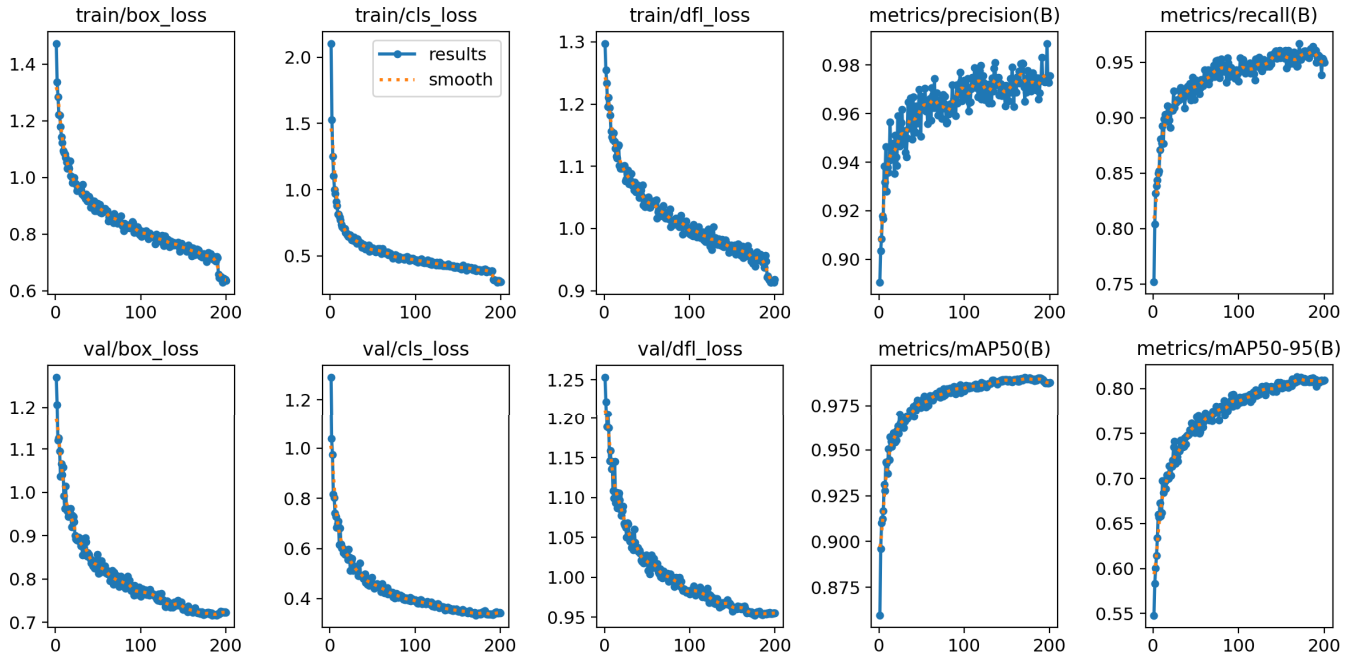


Fig. 6. Training and validation loss trends for CMSFF-YOLOv8.

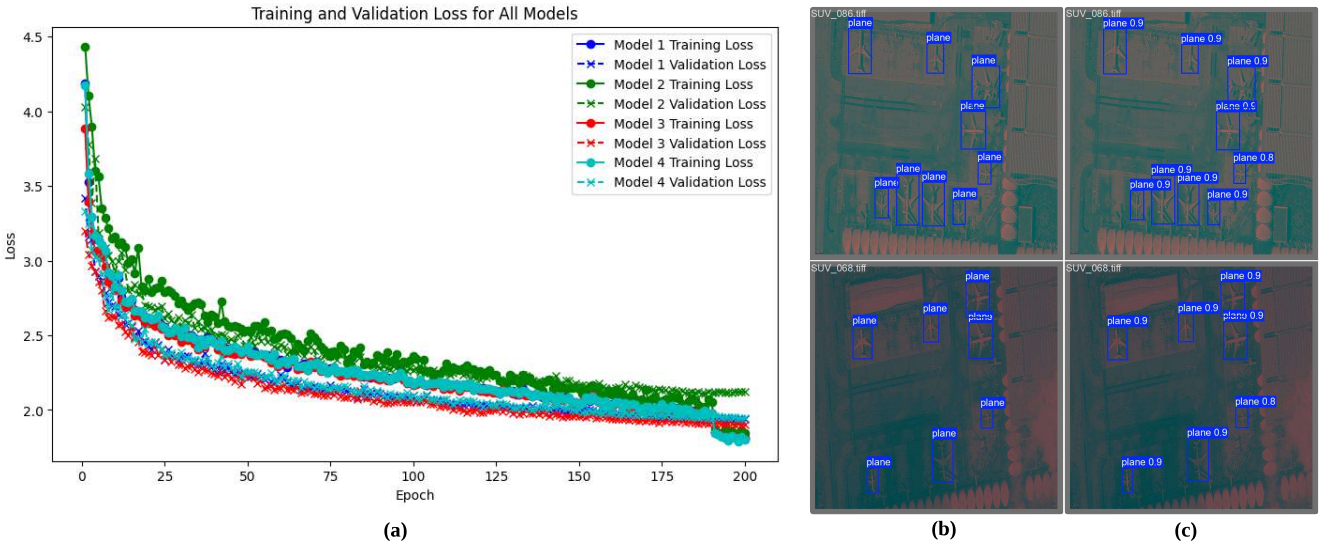


Fig. 7. Analysis of CMSFF-YOLOv8. (a) Training and validation loss trends for CMSFF-YOLOv8 architecture. (b) Plane detection only. (c) Plane detection with its model confidence score at the training and validation process.

and sophisticated preprocessing methods function, especially in the proprietary YOLOv8 architecture, which outperformed the other models in every important performance indicator.

In Fig. 6, the image shows several plots that illustrate the training and validation performance of the modified YOLOv8 object detection model across multiple epochs. The first set of plots represents the training losses: train/box$_{\text{loss}}$, train/cls$_{\text{loss}}$, and train/dfl$_{\text{loss}}$. These losses reflect the model's performance in predicting bounding boxes, classifying objects, and fine-tuning localization, respectively. The validation losses are represented by the second set of graphs val/box$_{\text{loss}}$, val/cls$_{\text{loss}}$, and val/dfl$_{\text{loss}}$. These plots show a similar pattern and aid in assessing how well the model generalizes to new data. In the second row, the performance metrics are shown, including metrics/precision, metrics/recall, and metrics/mAP50. These plots demonstrate the evolution of key performance measures during training.

Fig. 7(a) displays a graph of each model's training and validation losses across 200 epochs. To distinguish each model, its curves are color-coded. In Model 1, the training loss and validation loss are represented by blue dashed lines; in Model 2, the training loss and validation loss are represented by green dashed lines; in Model 3, the training loss and validation loss are represented by red dashed lines; and in

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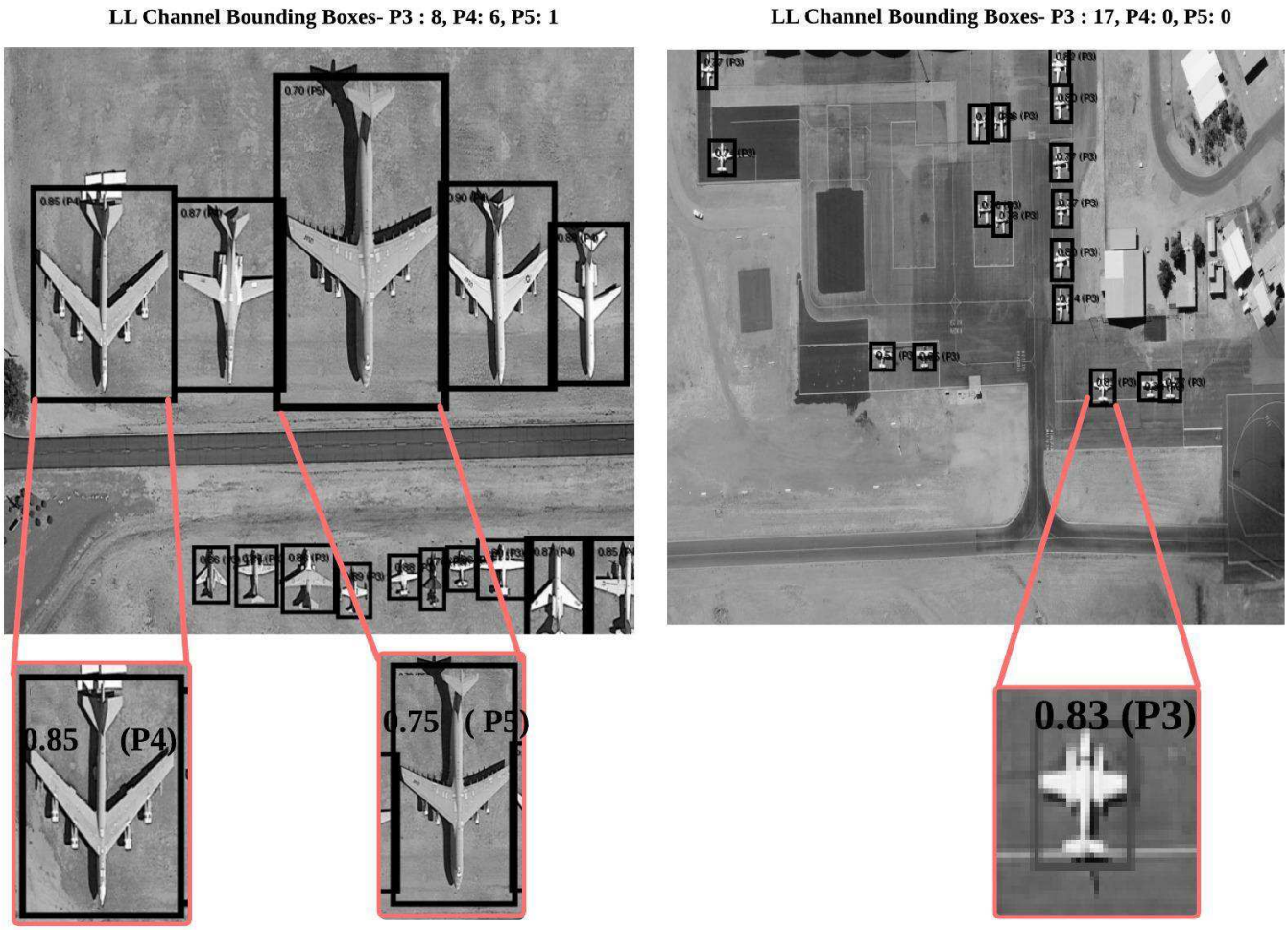


Fig. 8. Detection of planes in the test images, displaying confidence scores and categorizing detections based on three size scales: P3–P5 for better performance.

Model 4, the training loss and validation loss are represented by cyan dashed lines. As the number of epochs increases, all models exhibit a discernible drop in both training and validation loss, which is characteristic of models that converge during training.

The training and validation results of an object identification model are shown in Fig. 7(b), which is an image displaying predicted classes within the boxes around the plane. These bounding boxes indicate the positions of the planes detected by the model in the satellite image taken during the validation phase. However, this image does not include confidence ratings. Fig. 7(c) shows a confidence rating of "plane 0.9." The model also provides confidence scores that give insight into its certainty level for each prediction. The model performed its prediction when given an unknown image-mapped model; predictions with objects were analyzed and are displayed in Fig. 8.

The boxes show where the model placed planes, with the relevant parts of the class name accompanied by the confidence score of that prediction (e.g., "plane 0.87"). This means that the higher the value, the more certain the model is about the items that it plans to detect. The coordinates of the bounding box, as well as the class labels, will be included in the annotation file that the model writes in the YOLO format. Alongside the visual bounding boxes from the image, the generated annotation file assists with postprocessing tasks such as computing detection metrics, performing additional training, and fine-tuning the model for improved performance in subsequent rounds.

V. Conclusion

In the proposed method, a novel compressed-domain aircraft detection framework is introduced that integrates enhanced preprocessing techniques with a customized YOLOv8 architecture. By leveraging a 3-DWT, AHE, FDF, and an FT mechanism, we constructed a four-channel compressed image representation that enhances contrast, preserves high-frequency edge details, and reduces noise. Unlike conventional approaches that rely on the spatial domain, CMSFF-YOLOv8 performs detection directly on DWT subbands (LL3, LH3, HL3, and HH3). These processed four-channel images are fed into CMSFF-YOLOv8, which enhances feature extraction and improves detection accuracy by focusing on important regions. We designed custom architectural components tailored for frequency-domain processing, where SEConv is used to extract structural features from the LL3 subband and MGEFA is employed to capture edge-aware details from the high-frequency subbands. The CMSFF-YOLOv8 compares high-frequency edge details processed through the MGEFA

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module with the structural information from the LL subband processed through SEConv. Enhanced performance is achieved by modifying the YOLOv8 backbone using modules such as SFFN, FEFM, and FFM, which enable MSFF and improve the detection of small objects.

Our experiments on the HRPlanes, RarePlanes, and iSAID datasets show that the CMSFF-YOLOv8 model outshines the others, achieving the highest mAP of 0.90895 at 0.5, a precision of 0.87460, a recall of 0.82165, and an $F1$-score of 0.847299, which validate the effectiveness of our approach. Specifically, the mixed dataset experiment (HRPlanes, RarePlanes, and iSAID) confirms that our model generalizes well, with an $F1$-score of 0.8626 and mAP@0.5 of 0.9015, further validating its robustness in diverse remote sensing scenarios. In critical situations where the identification of small planes is essential, our model delivered superior identification results compared to existing models. This method bridges the gap between real-time detection and raw satellite image processing through compression domain detection, offering a scalable and computationally effective solution. It offers practical advantages for real-time applications in military surveillance, airport security, disaster response, and aerial traffic monitoring. Potential avenues for further research include integrating attention processes in feature extraction with deep learning networks, which can further enhance detection performance.

REFERENCES

[1] A. Singh, S. Kumar, and D. Choudhury, "Deep learning driven real-time airspace monitoring using satellite imagery," *Scalable Comput., Pract. Exper.*, vol. 25, no. 6, pp. 5672–5687, Oct. 2024.

[2] S. Majumdar, "The role of remote sensing and GIS in military strategy to prevent terror attacks," in *Intelligent Data Analytics for Terror Threat Prediction: Architectures, Methodologies, Techniques, and Applications*, 2021, pp. 79–94.

[3] S. Yıldız, O. Aydemir, A. Memiş, and S. Varlı, "A turnaround control system to automatically detect and monitor the time stamps of ground service actions in airports: A deep learning and computer vision based approach," *Eng. Appl. Artif. Intell.*, vol. 114, Sep. 2022, Art. no. 105032.

[4] G. Cheng and J. Han, "A survey on object detection in optical remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 117, pp. 11–28, Jul. 2016.

[5] S. Li, W. Song, L. Fang, Y. Chen, P. Ghamisi, and J. A. Benediktsson, "Deep learning for hyperspectral image classification: An overview," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6690–6709, Sep. 2019.

[6] J. Tang, C. Deng, G.-B. Huang, and B. Zhao, "Compressed-domain ship detection on spaceborne optical image using deep neural network and extreme learning machine," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 3, pp. 1174–1185, Mar. 2015.

[7] K. Liu and G. Mattyus, "Fast multiclass vehicle detection on aerial images," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 9, pp. 1938–1942, Sep. 2015.

[8] Y. Zhang, M. Ye, G. Zhu, Y. Liu, P. Guo, and J. Yan, "FFCA-YOLO for small object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5611215.

[9] M. Jindal, N. Raj, and P. Saranya, "Aircraft detection from remote sensing images using YOLOV5 architecture," in *Proc. 6th Int. Conf. Devices, Circuits Syst. (ICDCS)*, Apr. 2022, pp. 332–336.

[10] Z. Liu, Y. Gao, and Q. Du, "YOLO-class: Detection and classification of aircraft targets in satellite remote sensing images based on YOLO-extract," *IEEE Access*, vol. 11, pp. 109179–109188, 2023.

[11] Z. Liu, Y. Gao, Q. Du, M. Chen, and W. Lv, "YOLO-extract: Improved YOLOv5 for aircraft object detection in remote sensing images," *IEEE Access*, vol. 11, pp. 1742–1751, 2023.

[12] K. Zhao and X. Ren, "Small aircraft detection in remote sensing images based on YOLOv3," *IOP Conf. Ser., Mater. Sci. Eng.*, vol. 533, no. 1, May 2019, Art. no. 012056.

[13] N. Chaibi and M. Zaied, "A multi-level wavelet decomposition network for image super resolution," in *Proc. Int. Conf. Intell. Syst. Design Appl.* Cham, Switzerland: Springer, 2023, pp. 438–444.

[14] J. Wang, Z. Wang, J. Li, and J. Wu, "Multilevel wavelet decomposition network for interpretable time series analysis," in *Proc. 24th ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining*, Jul. 2018, pp. 2437–2446.

[15] B. E. Carvajal-Gámez, A. E. Moreno-Cervantes, F. J. Gallegos-Funes, and V. Ponomaryov, "Image reconstruction in complex media by the fourth moment wavelet," in *Proc. 9th Int. Kharkiv Symp. Phys. Eng. Microw., Millim. Submillimeter Waves (MSMW)*, Jun. 2016, pp. 1–3.

[16] S. Praveen kumar and S. Sridevi, "Image fusion algorithm for medical images using DWT and SR," in *Proc. Int. Conf. Artif. Intell. Smart Syst. (ICAIS)*, Mar. 2021, pp. 853–858.

[17] A. Ghasemzadeh and H. Demirel, "3D discrete wavelet transform-based feature extraction for hyperspectral face recognition," *IET Biometrics*, vol. 7, no. 1, pp. 49–55, Jan. 2018.

[18] D. McIntosh, T. P. Marques, and A. B. Albu, "Preservation of high frequency content for deep learning-based medical image classification," in *Proc. 18th Conf. Robots Vis. (CRV)*, May 2021, pp. 41–48.

[19] K. M. Aljebory, M. J. Yashar, and T. S. Mohammed, "Classification of EMG signals: Using DWT features and ANN classifier," *IAENG Int. J. Comput. Sci.*, vol. 51, no. 1, pp. 23–31, Jan. 2024.

[20] P. P. Khairnar and C. A. Manjare, "Image resolution and contrast enhancement of satellite geographical images with removal of noise using wavelet transforms," 2014, *arXiv:1405.1967*.

[21] M. S. Lazar and L. T. Bruton, "Combining the discrete wavelet transform and mixed-domain filtering," *IEEE Trans. Image Process.*, vol. 5, no. 7, pp. 1124–1136, Jul. 1996.

[22] J. Lee, S. R. Pant, and H.-S. Lee, "An adaptive histogram equalization based local technique for contrast preserving image enhancement," *Int. J. Fuzzy Log. Intell. Syst.*, vol. 15, no. 1, pp. 35–44, Mar. 2015.

[23] K. A. Dar and S. Mittal, "An enhanced adaptive histogram equalization based local contrast preserving technique for HDR images," *IOP Conf. Ser., Mater. Sci. Eng.*, vol. 1022, no. 1, Jan. 2021, Art. no. 012119.

[24] T. Wu and Y. Dong, "YOLO-SE: Improved YOLOv8 for remote sensing object detection and recognition," *Appl. Sci.*, vol. 13, no. 24, p. 12977, Dec. 2023.

[25] B. Yan, J. Li, Z. Yang, X. Zhang, and X. Hao, "AIE-YOLO: Auxiliary information enhanced YOLO for small object detection," *Sensors*, vol. 22, no. 21, p. 8221, Oct. 2022.

[26] J. Redmon and A. Farhadi, "YOLO9000: Better, faster, stronger," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 6517–6525.

[27] Q. Zhou, Z. Wang, Y. Zhong, F. Zhong, and L. Wang, "Efficient optimized YOLOv8 model with extended vision," *Sensors*, vol. 24, no. 20, p. 6506, Oct. 2024.

[28] A. Bochkovskiy, C.-Y. Wang, and H.-Y. Mark Liao, "YOLOv4: Optimal speed and accuracy of object detection," 2020, *arXiv:2004.10934*.

[29] Y. Li, Z. Liu, J. Yang, and H. Zhang, "Wavelet transform feature enhancement for semantic segmentation of remote sensing images," *Remote Sens.*, vol. 15, no. 24, p. 5644, Dec. 2023.

[30] S. Z. Mahmoodabadi, A. Ahmadian, M. D. Abolhasani, M. Eslami, and J. H. Bidgoli, "ECG feature extraction based on multiresolution wavelet transform," in *Proc. 27th Annu. Conf. IEEE Eng. Med. Biol.*, Mar. 2005, pp. 3902–3905.

[31] H. Huang, X. Tang, F. Wen, and X. Jin, "Small object detection method with shallow feature fusion network for chip surface defect detection," *Sci. Rep.*, vol. 12, no. 1, p. 3914, Mar. 2022.

[32] C. Song, Y. Li, and B. Huang, "A GPU-accelerated wavelet decompression system with SPIHT and Reed–Solomon decoding for satellite images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 4, no. 3, pp. 683–690, Sep. 2011.

[33] H. Fu, J. Liang, Z. Fang, J. Han, F. Liang, and G. Zhang, "WeConvene: Learned image compression with wavelet-domain convolution and entropy model," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2024, pp. 37–53.

[34] N. I. Chervyakov, P. A. Lyakhov, and N. N. Nagornov, "Quantization noise of multilevel discrete wavelet transform filters in image processing," *Optoelectron., Instrum. Data Process.*, vol. 54, no. 6, pp. 608–616, Nov. 2018.

[35] Y. Cai, Y. Zhao, S. Wen, and J. Feng, "Improved YOLOv8 SAR image aircraft object detection method," in *Proc. 7th Int. Symp. Auto. Syst. (ISAS)*, May 2024, pp. 1–6.

[36] J. A. Stark, "Adaptive image contrast enhancement using generalizations of histogram equalization," *IEEE Trans. Image Process.*, vol. 9, no. 5, pp. 889–896, May 2000.

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[37] X. Tan, X. Leng, R. Luo, Z. Sun, K. Ji, and G. Kuang, "YOLO-RC: SAR ship detection guided by characteristics of range-compressed domain," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 18834–18851, 2024.
[38] A. Mahmood, S. A. Khan, S. Hussain, and E. M. Almaghayreh, "An adaptive image contrast enhancement technique for low-contrast images," *IEEE Access*, vol. 7, pp. 161584–161593, 2019.
[39] P. Härtinger and C. Steger, "Adaptive histogram equalization in constant time," *J. Real-Time Image Process.*, vol. 21, no. 3, p. 93, Jun. 2024.
[40] D. Unsal, "HRPlanesv2—High resolution satellite imagery for aircraft detection," *Remote Sens.*, Feb. 2022.
[41] J. Shermeyer, T. Hossler, A. V. Etten, D. Hogan, R. Lewis, and D. Kim, "RarePlanes: Synthetic data takes flight," in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2021, pp. 207–217.
[42] S. W. Zamir et al., "iSAID: A large-scale dataset for instance segmentation in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops*, Jul. 2019, pp. 28–37.
[43] T. Bakirman and E. Sertel, "A benchmark dataset for deep learning-based airplane detection: HRPlanes," 2022, *arXiv:2204.10959*.
[44] S. El Ghazouali, A. Gucciardi, N. Venturi, M. Rueegsegger, and U. Michelucci, "FlightScope: An experimental comparative review of aircraft detection algorithms in satellite imagery," 2024, *arXiv:2404.02877*.
[45] M. Tasyurek and E. Gul, "A new deep learning approach based on grayscale conversion and DWT for object detection on adversarial attacked images," *J. Supercomput.*, vol. 79, no. 18, pp. 20383–20416, Dec. 2023.
[46] C. Yang, Y. Zhao, and S. Wang, "Deep image compression in the wavelet transform domain based on high frequency sub-band prediction," *IEEE Access*, vol. 7, pp. 52484–52497, 2019.



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